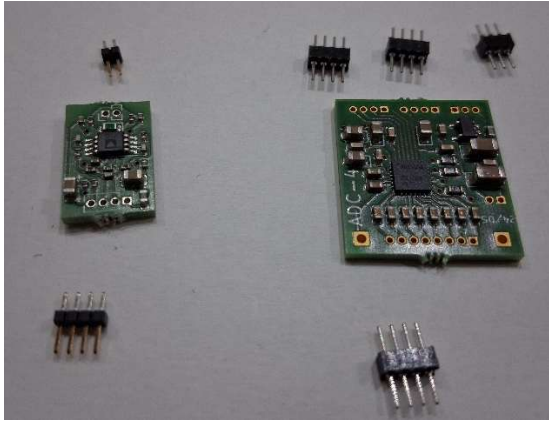


Recorder electronics

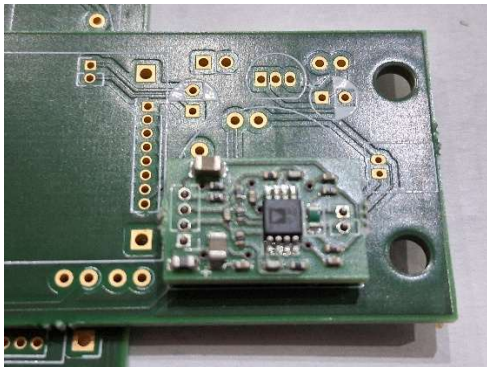
Electronic details

If wanted solder pin-headers to preamp and ADC boards



Using the base board to keep pin-headers straight, but without soldering to base board.

First, preamp

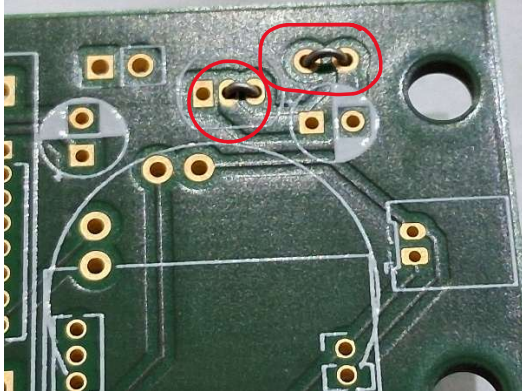


Then ADC board

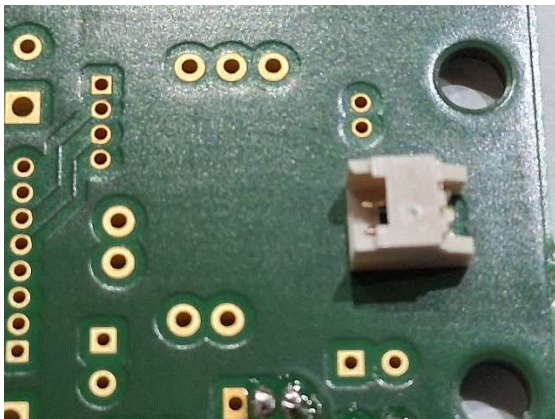


As we only have connections for the first two audio channels, a 4-pin header is sufficient. The preamp uses pins 3,4, while the preamp of Mark (not shown here) would use, pins 1 and 2.

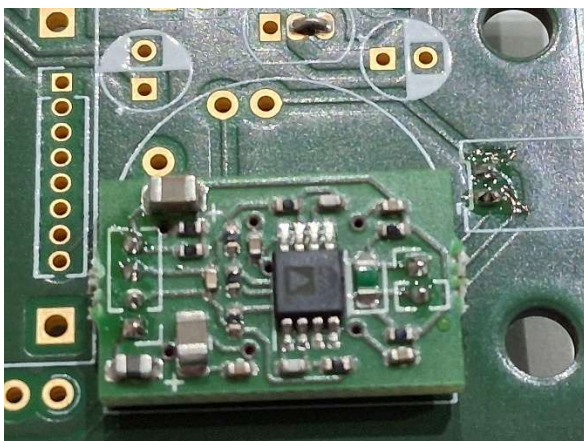
Connect the pins with some wire or solder as shown in next figure



Solder hydrophone header (here pico blade). To have free room for manoeuvres. Header is soldered on the back side as shown in figure.



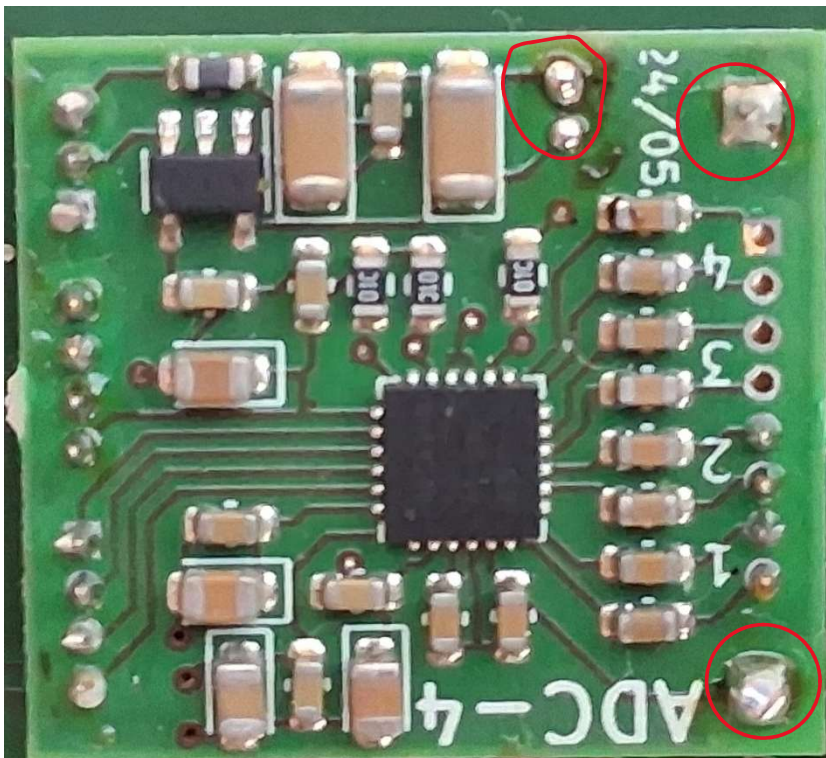
Solder preamp to base board



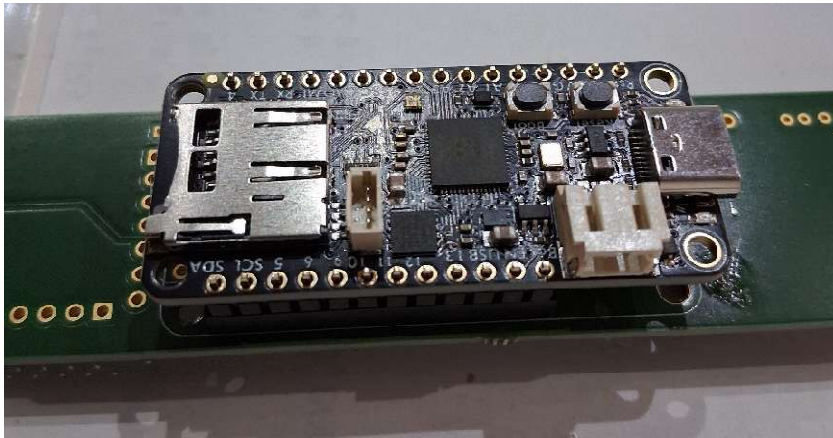
Solder ADC to base board (only four pins to the right are needed)



Solder 4 wires to additional connect ADC to base board



Solder pin header to Adalogger using short headers (short pin and receptacles)



Solder female header from other side to board (having Adalogger still attached)

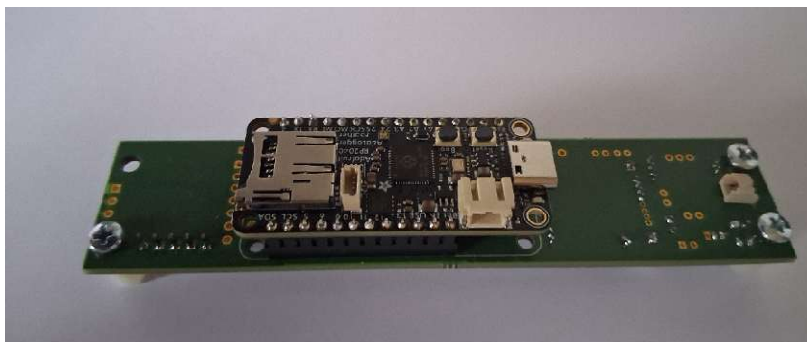
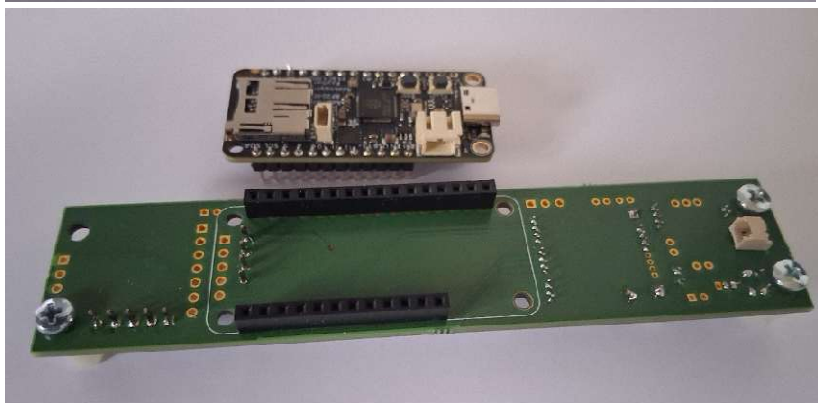
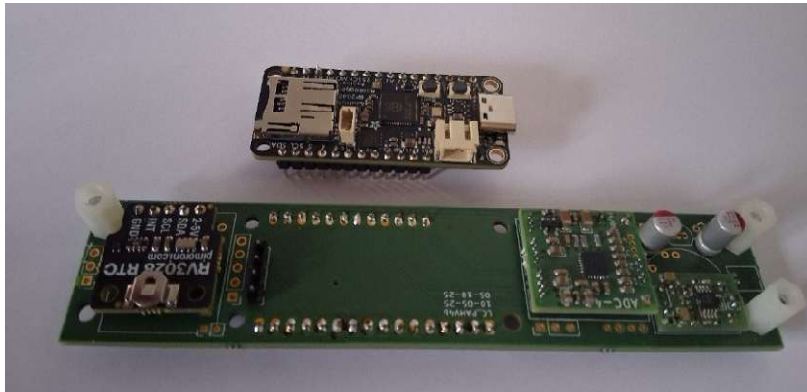


Remove Adalogger (for stability) and solder RTC to pin header. Make using the provided receptacle to keep board at 90°. Flip base board and solder pins to board.

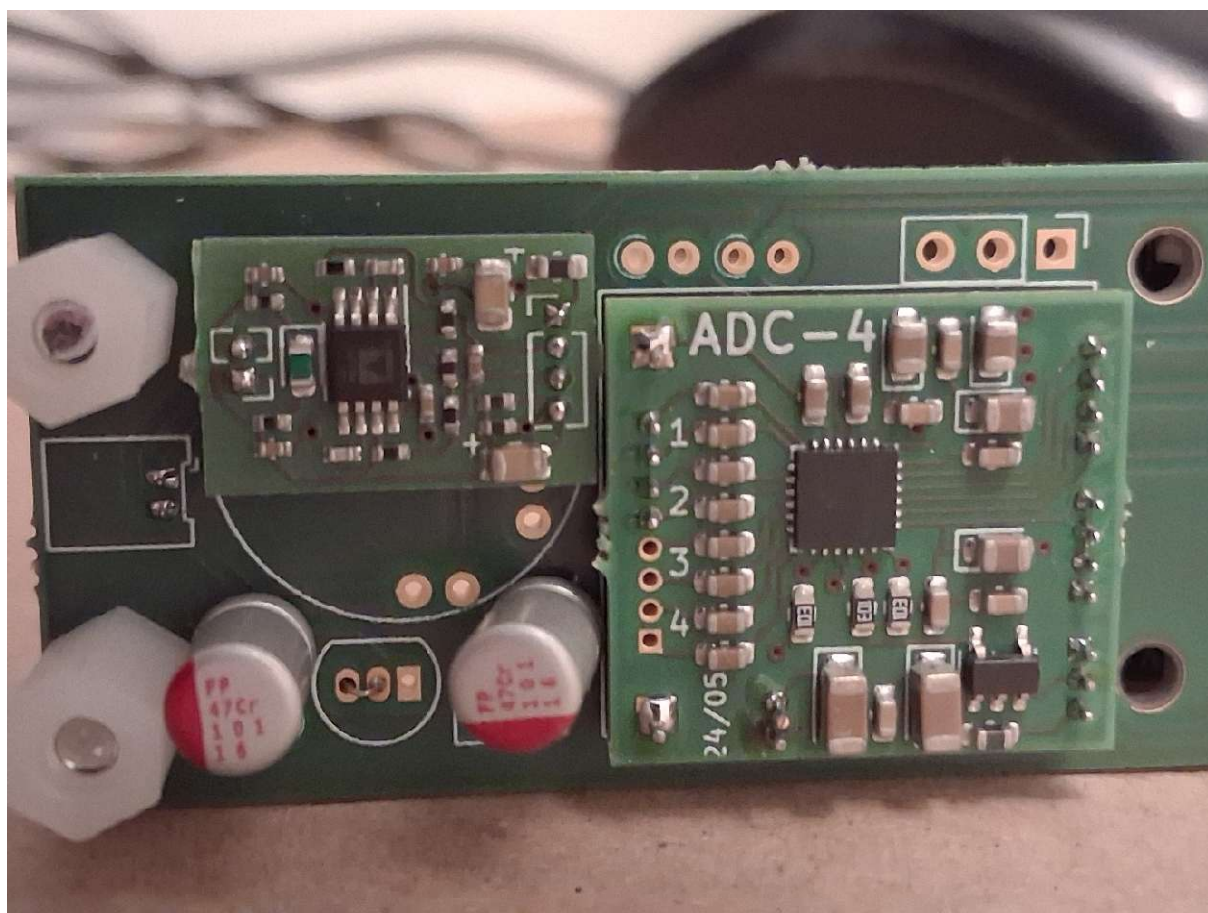


If needed: solder DC/DC converter with similar procedure as RTC clock (not shown in figure).

If needed: solder ELCOs

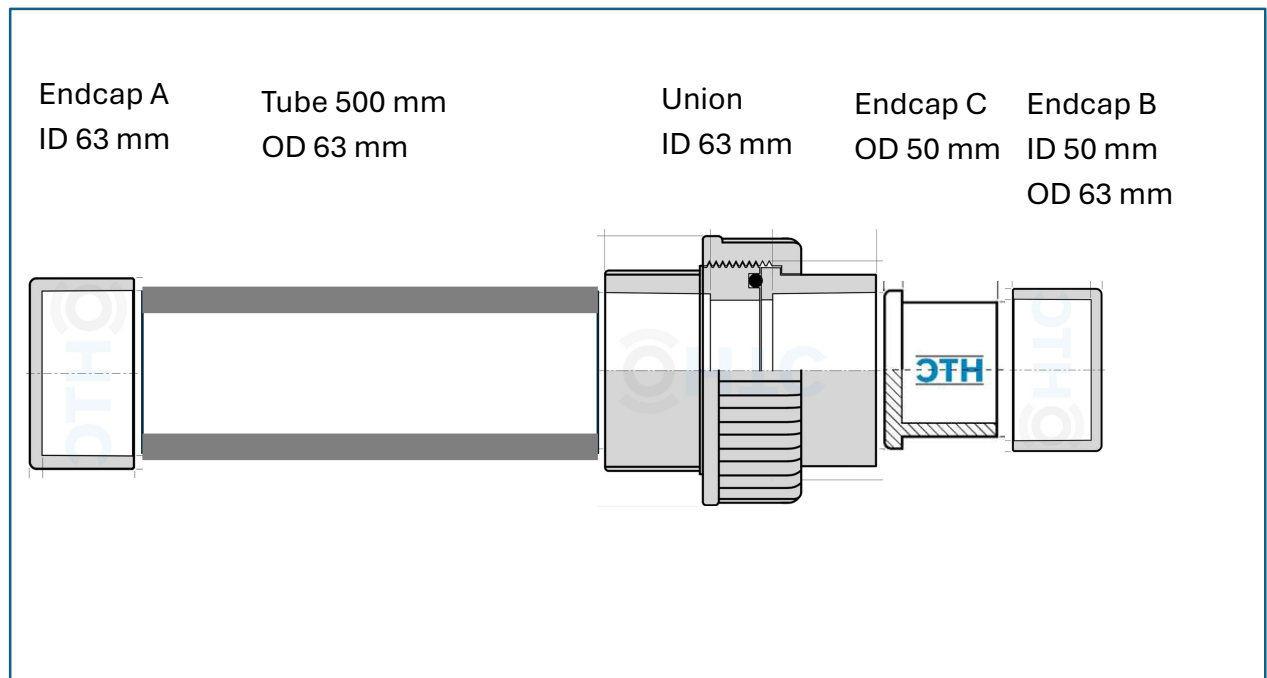


Close-up of Pre-amp ADC



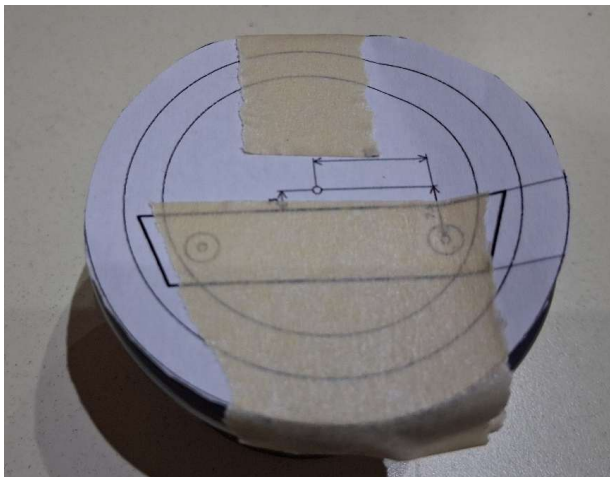
Pressure Container

Following instructions are for this configuration:



Endcaps

Take 63 mm OD flat endcap B, drill 3 mm hole in center



Take endcap (endcap C) with 63 mm top and 50 mm tube, drill 3mm holes also at center



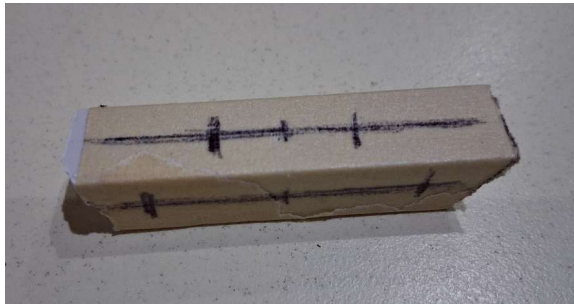
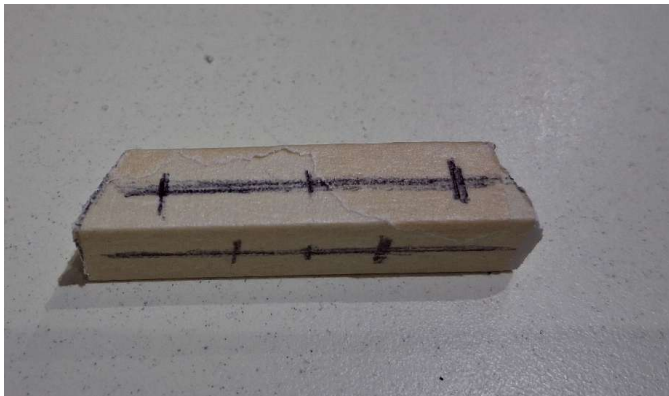
Mounting bar

Cut piece from 10 x 10 mm bar



D

Mark holes to drill (wider correspond to spacings of drawing used for center holes).



The narrower holes are separated by half the separation of the wider holes

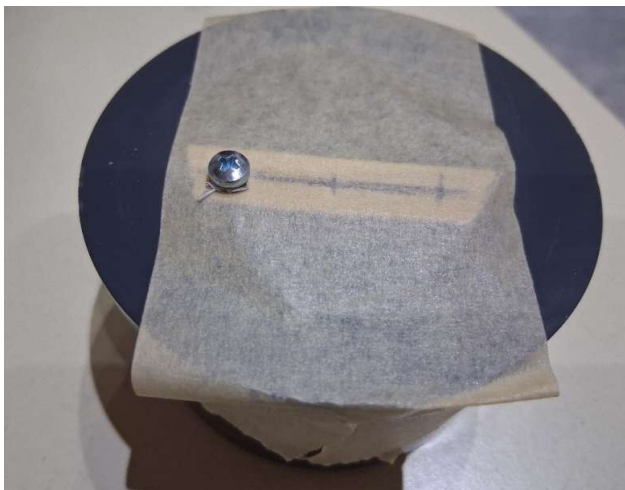
Combine mounting bar with Endcap combi

Insert into endcap combi



Fix with tape

Drill one 3 mm hole through bar and Endcap C and fix with screw before drilling the



other hole

Widen holes in bar to 4 / 4.5 mm

Resulting endcaps and holes



Adjust endcap combi for hydrophone and mounting bar

Widen the hole in the 63 mm OD endcap to 9 mm using tool to be perfectly vertical



Cut 10 mm thread into hole of 63 mm endcap



Cut 4-mm thread in the off-center holes in 50 mm OD Endcap

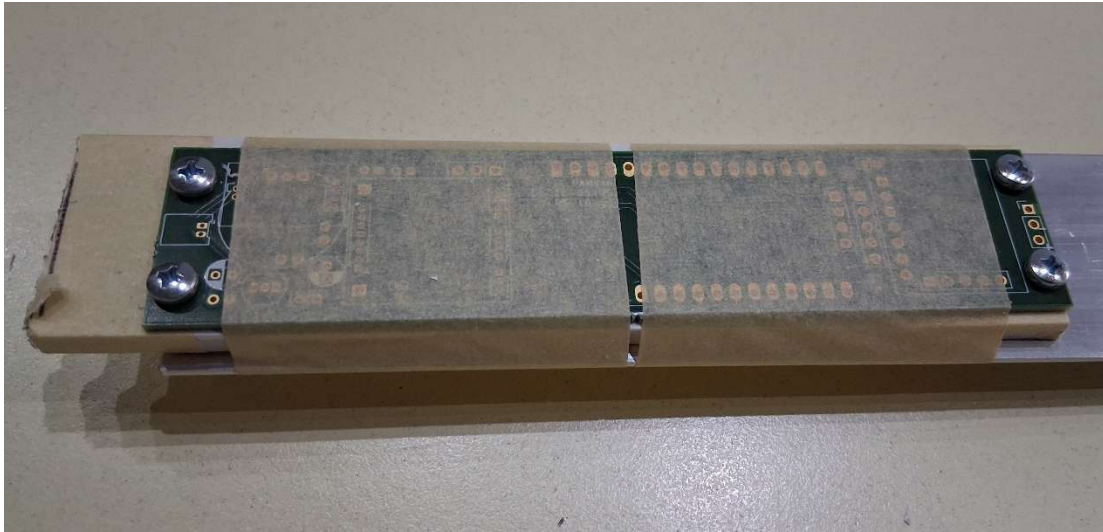




Widen center hole in 50 mm OD endcap to about 24 mm diameter

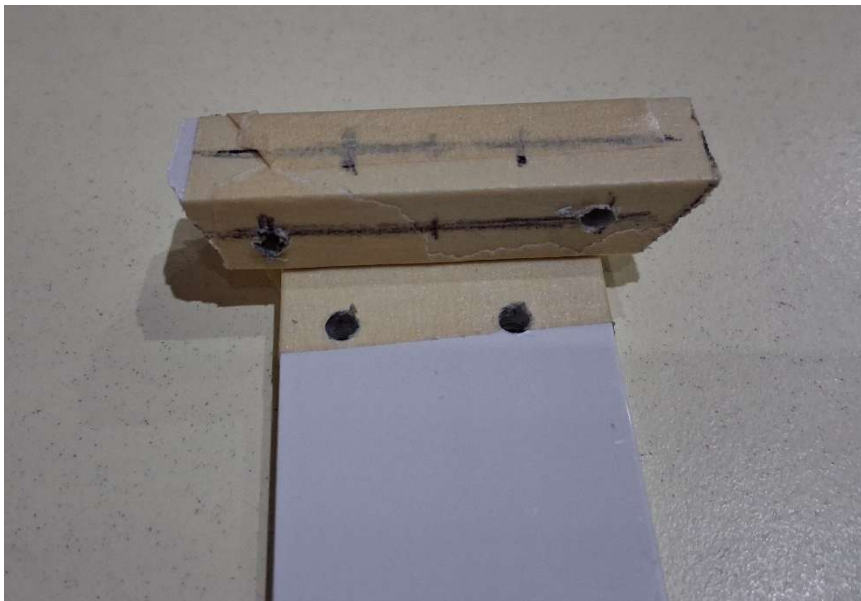
Combine electronics to flat pvc and alu bar

Tape board or stencil om pvc bar and alu bar

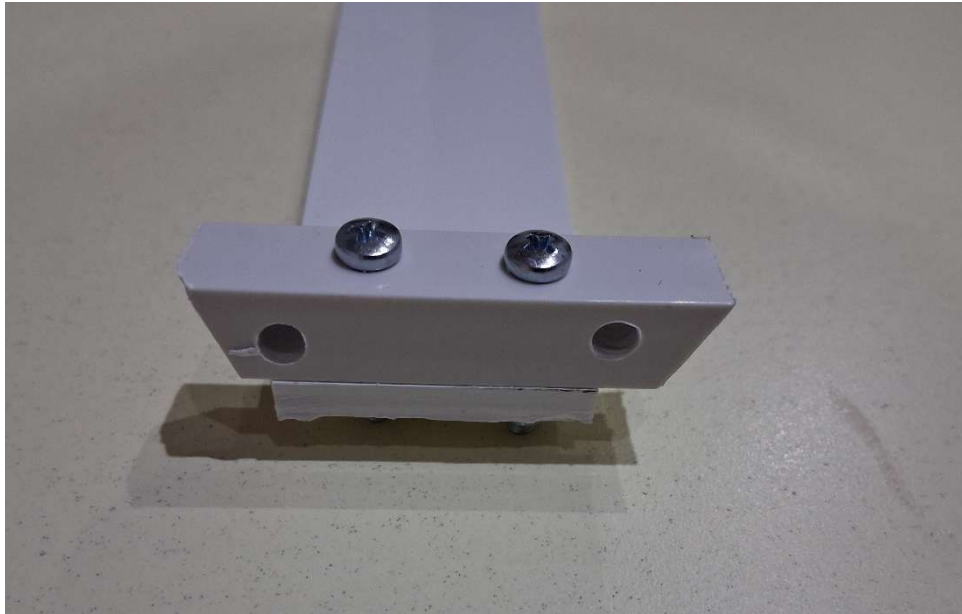


Drill 4 holes through pvc and alu plate

place 10x10 mm bar on flat PVC bar, smaller side touching plate



drill two 3 mm holes through bar and plate, fix again with 3mm bolt after first hole

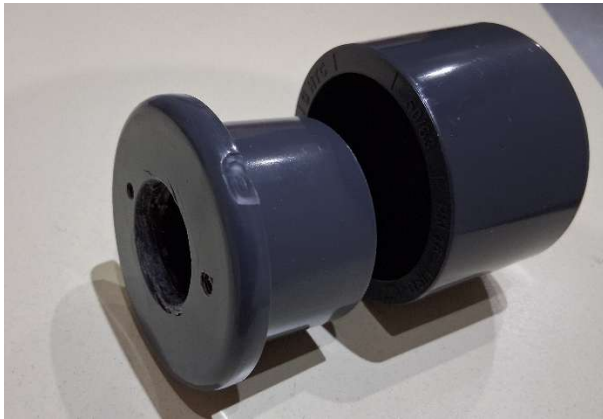


Fix pvc plate to alu plate with 12 mm screws and three 10 mm nylon standoff and one nut

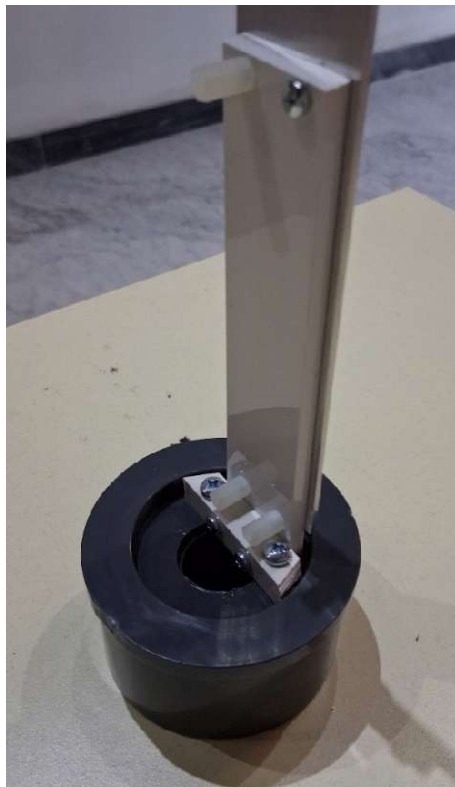


Assemble recorder

Press mounting cap (counter end-cap) into hydrophone end-cap



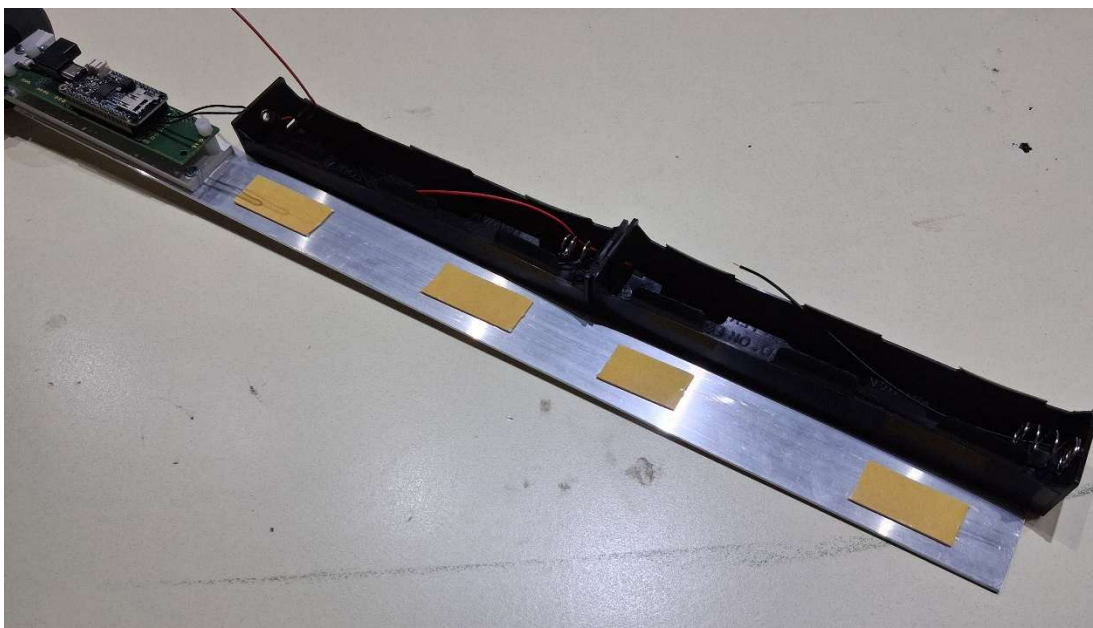
Assemble mounting plate to endcap combi



Adding recorder



Place battery packs long side alu plate, place 4 double sided tapes on alu plate and fix battery packs, keeping 1 mm distance from pvc plate. Hydrophone holders may extend over 50 cm alu bar.



Alternative mounting methods

Au U with 10 mm hex nut attached directly to hydrophone penetrator (here with Teensy 4.1) See below.



PVC bar screwed onto counter end-cap



3D printed insert replacing counter end-cap



Construction of Alu bracket

Take an Alu U profile (20 x 10 mm)



mark a point in the middle and about 16-17 mm from the end.

Drill a 3 mm hole as precisely as possible at this point

Drill next a 10 mm hole using for example a drill head as shown in picture



Cut the Alu U profile such that the 10 mm hole is in the middle.



Fix the Alu U profile on top of the plastic bar (were the electronics will be placed)



Drill from below two 3 mm holes that are about 5 to 6 mm from the end and 4 mm from the side-

Fix the Alu profile to the plastic bar with M3 bolts.

Gluing PV

Take endcaps and unions

Final Recorder

Glue endcaps and unions together onto tube

